ARCTIC OR POLAR CLIMATE

POLAR CLIMATE WORLD DISTRIBUTION → TEMPERATURE-PRECIPITATION PROFILE → TUNDRA VEGETATION BELT → ARCTIC-ANTARCTIC ASYMMETRY → ICE-ALBEDO FEEDBACK LOOP → PERMAFROST CARBON FEEDBACK

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g5 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 POLAR CLIMATE WORLD DISTRIBUTION

POLAR CLIMATE WORLD DISTRIBUTION | GREENLAND AND HIGH-LATITUDE HIGHLANDS, PERMANENT SNOW COVER | COASTAL GREENLAND, N. CANADA, ALASKA, EURASIAN ARCTIC SEABOARD | SOUTHERN POLE | CONTINENT UNDER ICE, NO INDIGENOUS POPULATION | NORTHERN POLE | OCEAN BASIN SURROUNDED BY CONTINENTAL LAND MASSES | MUST REMEMBER

ICE-CAP → Greenland and high-latitude highlands, permanent snow cover

TUNDRA → coastal Greenland, N. Canada, Alaska, Eurasian Arctic seaboard → SOUTHERN POLE → Antarctica: continent under ice, no indigenous population → NORTHERN POLE → Arctic: ocean basin surrounded by continental land masses

CORE CLAIM
- ICE-CAP → Greenland and high-latitude highlands, permanent snow cover
- TUNDRA → coastal Greenland, N. Canada, Alaska, Eurasian Arctic seaboard

MECHANISM
- SOUTHERN POLE → Antarctica: continent under ice, no indigenous population
- NORTHERN POLE → Arctic: ocean basin surrounded by continental land masses

CONSEQUENCE / LIMIT
- MUST REMEMBER: Polar climates separate tundra ET, where the warmest month is above 0°C but below 10°C, from ice-cap EF below 0°C year-round
- low sun angle, high albedo, polar night/day, permafrost and short food webs structure Arctic and Antarctic systems.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Polar climates separate tundra ET, where the warmest month is above 0°C but below 10°C, from ice-cap EF below 0°C year-round; low sun angle, high albedo, polar night/day, permafrost and short food webs structure Arctic and Antarctic systems.

01

STAGE 01 TEMPERATURE-PRECIPITATION PROFILE

TEMPERATURE-PRECIPITATION PROFILE | WARMEST MONTH | SELDOM EXCEEDS ABOUT 10 DEGREES C (JUNE/JULY) | MONTHS ABOVE FREEZING | NO MORE THAN ABOUT FOUR PER YEAR | LOW, MOSTLY SNOW; POLAR CLIMATE | COLD DESERT | WIND REDISTRIBUTING EXISTING SNOW, NOT HEAVY SNOWFALL EVENTS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
WARMEST MONTH → seldom exceeds about 10 degrees C (June/July)	PRECIPITATION → low, mostly snow; polar climate = cold desert	WARMEST MONTH → seldom exceeds about 10 degrees C (June/July)
MONTHS ABOVE FREEZING → no more than about four per year	BLIZZARDS → wind redistributing existing snow, not heavy snowfall events	MONTHS ABOVE FREEZING → no more than about four per year

CORE CLAIM
- WARMEST MONTH → seldom exceeds about 10 degrees C (June/July)
- MONTHS ABOVE FREEZING → no more than about four per year

MECHANISM
- PRECIPITATION → low, mostly snow; polar climate = cold desert
- BLIZZARDS → wind redistributing existing snow, not heavy snowfall events

CONSEQUENCE / LIMIT
- WARMEST MONTH → seldom exceeds about 10 degrees C (June/July)
- MONTHS ABOVE FREEZING → no more than about four per year

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BLIZZARDS → wind redistributing existing snow, not heavy snowfall events.

02

STAGE 02 TUNDRA VEGETATION BELT

TUNDRA VEGETATION BELT | PREVENTS DEEP ROOTING, IMPEDES DRAINAGE | SHORT GROWING SEASON | WARMEST MONTH BELOW 10 DEGREES C | STUNTED PLANTS | DWARF WILLOWS, BIRCHES, ALDERS IN SHELTERED SPOTS | LICHEN GROUND COVER | REINDEER MOSS PROVIDES GRAZING FOR REINDEER HERDS

PERMAFROST → prevents deep rooting, impedes drainage → SHORT GROWING SEASON → warmest month below 10 degrees C → STUNTED PLANTS → dwarf willows, birches, alders in sheltered spots → LICHEN GROUND COVER → reindeer moss provides grazing for reindeer herds

CORE CLAIM
- PERMAFROST → prevents deep rooting, impedes drainage
- SHORT GROWING SEASON → warmest month below 10 degrees C

MECHANISM
- STUNTED PLANTS → dwarf willows, birches, alders in sheltered spots
- LICHEN GROUND COVER → reindeer moss provides grazing for reindeer herds

CONSEQUENCE / LIMIT
- PERMAFROST → prevents deep rooting, impedes drainage
- SHORT GROWING SEASON → warmest month below 10 degrees C

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LICHEN GROUND COVER → reindeer moss provides grazing for reindeer herds.

03

STAGE 03 ARCTIC-ANTARCTIC ASYMMETRY

ARCTIC-ANTARCTIC ASYMMETRY | OCEAN SURROUNDED BY CONTINENTS; MAINLY FLOATING SEA ICE | CONTINENT SURROUNDED BY OCEAN; VAST GROUNDED ICE SHEET | ARCTIC AMPLIFICATION MUCH FASTER; ANTARCTIC MORE COMPLEX | SOVEREIGN STATES + FORUM | TREATY REGIME

CORE CLAIM
- PERMAFROST → prevents deep rooting, impedes drainage
- SHORT GROWING SEASON → warmest month below 10 degrees C

MECHANISM
- STUNTED PLANTS → dwarf willows, birches, alders in sheltered spots
- LICHEN GROUND COVER → reindeer moss provides grazing for reindeer herds

CONSEQUENCE / LIMIT
- PERMAFROST → prevents deep rooting, impedes drainage
- SHORT GROWING SEASON → warmest month below 10 degrees C

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LICHEN GROUND COVER → reindeer moss provides grazing for reindeer herds.

03

STAGE 03

ARCTIC-ANTARCTIC ASYMMETRY

ARCTIC-ANTARCTIC ASYMMETRY | OCEAN SURROUNDED BY CONTINENTS; MAINLY FLOATING SEA ICE | CONTINENT SURROUNDED BY OCEAN; VAST GROUNDED ICE SHEET | ARCTIC AMPLIFICATION MUCH FASTER; ANTARCTIC MORE COMPLEX | SOVEREIGN STATES + FORUM | TREATY REGIME

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
ARCTIC → ocean surrounded by continents; mainly floating sea ice	WARMING → Arctic amplification much faster; Antarctic more complex	Antarctic: treaty regime
ANTARCTIC → continent surrounded by ocean; vast grounded ice sheet	GOVERNANCE → Arctic: sovereign states + forum	ARCTIC → ocean surrounded by continents; mainly floating sea ice

CORE CLAIM
- ARCTIC → ocean surrounded by continents; mainly floating sea ice
- ANTARCTIC → continent surrounded by ocean; vast grounded ice sheet

MECHANISM
- WARMING → Arctic amplification much faster; Antarctic more complex
- GOVERNANCE → Arctic: sovereign states + forum

CONSEQUENCE / LIMIT
- Antarctic: treaty regime
- ARCTIC → ocean surrounded by continents; mainly floating sea ice

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GOVERNANCE → Arctic: sovereign states + forum | Antarctic: treaty regime.

04

STAGE 04

ICE-ALBEDO FEEDBACK LOOP

ICE-ALBEDO FEEDBACK LOOP | SEA ICE AND SNOW RETREAT | BRIGHT SURFACE REPLACED BY DARK OCEAN/GROUND | ALBEDO DROPS | FAR MORE SOLAR RADIATION ABSORBED INSTEAD OF REFLECTED | SURFACE WARMING | FURTHER ICE AND SNOW RETREAT (SELF-REINFORCING) | REINFORCING FACTORS

SEA ICE AND SNOW RETREAT → bright surface replaced by dark ocean/ground → ALBEDO DROPS → far more solar radiation absorbed instead of reflected → SURFACE WARMING → further ice and snow retreat (self-reinforcing) → REINFORCING FACTORS → thin polar atmosphere, ocean heat escape, poleward transport

CORE CLAIM
- SEA ICE AND SNOW RETREAT → bright surface replaced by dark ocean/ground
- ALBEDO DROPS → far more solar radiation absorbed instead of reflected

MECHANISM
- SURFACE WARMING → further ice and snow retreat (self-reinforcing)
- REINFORCING FACTORS → thin polar atmosphere, ocean heat escape, poleward transport

CONSEQUENCE / LIMIT
- SEA ICE AND SNOW RETREAT → bright surface replaced by dark ocean/ground
- ALBEDO DROPS → far more solar radiation absorbed instead of reflected

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REINFORCING FACTORS → thin polar atmosphere, ocean heat escape, poleward transport.

05

STAGE 05

PERMAFROST CARBON FEEDBACK

PERMAFROST CARBON FEEDBACK | THAWS FROZEN GROUND CONTAINING ANCIENT ORGANIC MATTER | DRY CONDITIONS | DECOMPOSITION RELEASES CARBON DIOXIDE | WATERLOGGED CONDITIONS | DECOMPOSITION RELEASES METHANE (STRONGER GHG) | RELEASED GHGS CAUSE FURTHER WARMING, FURTHER THAW

WARMING → thaws frozen ground containing ancient organic matter → DRY CONDITIONS → decomposition releases carbon dioxide → WATERLOGGED CONDITIONS → decomposition releases methane (stronger GHG) → FEEDBACK → released GHGs cause further warming, further thaw

CORE CLAIM
- WARMING → thaws frozen ground containing ancient organic matter
- DRY CONDITIONS → decomposition releases carbon dioxide

MECHANISM
- WATERLOGGED CONDITIONS → decomposition releases methane (stronger GHG)
- FEEDBACK → released GHGs cause further warming, further thaw

CONSEQUENCE / LIMIT
- WARMING → thaws frozen ground containing ancient organic matter
- DRY CONDITIONS → decomposition releases carbon dioxide

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FEEDBACK → released GHGs cause further warming, further thaw.

06

STAGE 06

SEA-LEVEL GROUNDED-ICE RULE

SEA-LEVEL GROUNDED-ICE RULE | FLOATING SEA ICE MELTS | SEA LEVEL DOES NOT RISE (ALREADY DISPLACING WATER) | GROUNDED ICE MELTS | MASS TRANSFERRED TO OCEAN, SEA LEVEL RISES | ICE SHELVES LOST | REMOVES BUTTRESSING, ACCELERATES GROUNDED-ICE DISCHARGE | EXAM RULE

FLOATING SEA ICE MELTS → sea level does NOT rise (already displacing water)

GROUNDED ICE MELTS → mass transferred to ocean, sea level RISES → ICE SHELVES LOST → removes buttressing, accelerates grounded-ice discharge → EXAM RULE → always distinguish floating from grounded in any polar answer

CORE CLAIM
- FLOATING SEA ICE MELTS → sea level does NOT rise (already displacing water)
- GROUNDED ICE MELTS → mass transferred to ocean, sea level RISES

MECHANISM
- ICE SHELVES LOST → removes buttressing, accelerates grounded-ice discharge
- EXAM RULE → always distinguish floating from grounded in any polar answer

CONSEQUENCE / LIMIT
- FLOATING SEA ICE MELTS → sea level does NOT rise (already displacing water)
- GROUNDED ICE MELTS → mass transferred to ocean, sea level RISES

06

STAGE 06

SEA-LEVEL GROUNDED-ICE RULE

SEA-LEVEL GROUNDED-ICE RULE · FLOATING SEA ICE MELTS · SEA LEVEL DOES NOT RISE (ALREADY DISPLACING WATER) · GROUNDED ICE MELTS · MASS TRANSFERRED TO OCEAN, SEA LEVEL RISES · ICE SHELVES LOST · REMOVES BUTTRESSING, ACCELERATES GROUNDED-ICE DISCHARGE · EXAM RULE

FLOATING SEA ICE MELTS → sea level does NOT rise (already displacing water)

GROUNDED ICE MELTS → mass transferred to ocean, sea level RISES → ICE SHELVES LOST → removes buttressing, accelerates grounded-ice discharge → EXAM RULE → always distinguish floating from grounded in any polar answer

CORE CLAIM
- FLOATING SEA ICE MELTS → sea level does NOT rise (already displacing water)
- GROUNDED ICE MELTS → mass transferred to ocean, sea level RISES

MECHANISM
- ICE SHELVES LOST → removes buttressing, accelerates grounded-ice discharge
- EXAM RULE → always distinguish floating from grounded in any polar answer

CONSEQUENCE / LIMIT
- FLOATING SEA ICE MELTS → sea level does NOT rise (already displacing water)
- GROUNDED ICE MELTS → mass transferred to ocean, sea level RISES

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXAM RULE → always distinguish floating from grounded in any polar answer.

07

STAGE 07

ARCTIC SHIPPING CONSTRAINTS

ARCTIC SHIPPING CONSTRAINTS · ROUTES SHORTER THAN SUEZ/PANAMA BETWEEN NE ASIA AND NW · MOBILE ICE · UNPREDICTABLE, HAZARDOUS DESPITE RETREAT TREND · RESCUE AND CHARTING · THIN CAPACITY, INCOMPLETE SURVEYS · NEAR-TERM TRAFFIC · RESOURCE EXPORT AND DESTINATION, NOT TRANSIT CONTAINER

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
ADVANTAGE → routes shorter than Suez/Panama between NE Asia and NW Europe	RESCUE AND CHARTING → thin capacity, incomplete surveys	ADVANTAGE → routes shorter than Suez/Panama between NE Asia and NW Europe
MOBILE ICE → unpredictable, hazardous despite retreat trend	NEAR-TERM TRAFFIC → resource export and destination, not transit container	MOBILE ICE → unpredictable, hazardous despite retreat trend

CORE CLAIM
- ADVANTAGE → routes shorter than Suez/Panama between NE Asia and NW Europe
- MOBILE ICE → unpredictable, hazardous despite retreat trend

MECHANISM
- RESCUE AND CHARTING → thin capacity, incomplete surveys
- NEAR-TERM TRAFFIC → resource export and destination, not transit container

CONSEQUENCE / LIMIT
- ADVANTAGE → routes shorter than Suez/Panama between NE Asia and NW Europe
- MOBILE ICE → unpredictable, hazardous despite retreat trend

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NEAR-TERM TRAFFIC → resource export and destination, not transit container.

08

STAGE 08

INDIA COLD DESERT LADAKH

INDIA COLD DESERT LADAKH · GREATER HIMALAYA BLOCKS SW MONSOON, TRANS-HIMALAYA STAYS ARID · HIGH ALTITUDE · INTENSE INSOLATION BY DAY, SHARP RADIATIVE COOLING BY NIGHT · WINTER PRECIPITATION · LIGHT SNOW FROM WESTERN DISTURBANCES · BIOME CLASS · COLD DESERT, SAME AS SIERRA NEVADA (USA) AND ANDES

RAIN-SHADOW → Greater Himalaya blocks SW monsoon, Trans-Himalaya stays arid

HIGH ALTITUDE → intense insolation by day, sharp radiative cooling by night → WINTER PRECIPITATION → light snow from western disturbances → BIOME CLASS → cold desert, same as Sierra Nevada (USA) and Andes (Argentina)

CORE CLAIM
- RAIN-SHADOW → Greater Himalaya blocks SW monsoon, Trans-Himalaya stays arid
- HIGH ALTITUDE → intense insolation by day, sharp radiative cooling by night

MECHANISM
- WINTER PRECIPITATION → light snow from western disturbances
- BIOME CLASS → cold desert, same as Sierra Nevada (USA) and Andes (Argentina)

CONSEQUENCE / LIMIT
- RAIN-SHADOW → Greater Himalaya blocks SW monsoon, Trans-Himalaya stays arid
- HIGH ALTITUDE → intense insolation by day, sharp radiative cooling by night

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BIOME CLASS → cold desert, same as Sierra Nevada (USA) and Andes (Argentina).

09

STAGE 09

INDIA POLAR STATIONS

INDIA POLAR STATIONS · DAKSHIN GANGOTRI (1983) · FIRST STATION, NOW DECOMMISSIONED/BURIED IN ICE · MAITRI (1989) · SCHIRMACHER OASIS, ANTARCTICA; OPERATIONAL · BHARATI (2012) · LARSEMANN HILLS, ANTARCTICA; OPERATIONAL · HIMADRI (2008)

DAKSHIN GANGOTRI (1983) → first station, now decommissioned/buried in ice

MAITRI (1989) → Schirmacher Oasis, Antarctica; operational → BHARATI (2012) → Larsemann Hills, Antarctica; operational → HIMADRI (2008) → Ny-Alesund, Svalbard; Arctic; HIMANSH → Spiti, HP

CORE CLAIM
- DAKSHIN GANGOTRI (1983) → first station, now decommissioned/buried in ice
- MAITRI (1989) → Schirmacher Oasis, Antarctica; operational

MECHANISM
- BHARATI (2012) → Larsemann Hills, Antarctica; operational
- HIMADRI (2008) → Ny-Alesund, Svalbard; Arctic; HIMANSH → Spiti, HP

CONSEQUENCE / LIMIT
- CLOSE DISTINCTION: Arctic is mainly an ocean surrounded by continents while Antarctica is a high continent surrounded by ocean
- sea ice differs from land ice, tundra from ice cap, and polar desert refers to low precipitation rather than absence of frozen water

09

STAGE 09 INDIA POLAR STATIONS

INDIA POLAR STATIONS · DAKSHIN GANGOTRI (1983) · FIRST STATION, NOW DECOMMISSIONED/BURIED IN ICE · MAITRI (1989) · SCHIRMACHER OASIS, ANTARCTICA; OPERATIONAL · BHARATI (2012) · LARSEMANN HILLS, ANTARCTICA; OPERATIONAL · HIMADRI (2008)

DAKSHIN GANGOTRI (1983) → first station, now decommissioned/buried in ice

MAITRI (1989) → Schirmacher Oasis, Antarctica; operational → BHARATI (2012) → Larsemann Hills, Antarctica; operational → HIMADRI (2008) → Ny-Alesund, Svalbard; Arctic; HIMANSH → Spiti, HP

CORE CLAIM
- DAKSHIN GANGOTRI (1983) → first station, now decommissioned/buried in ice
- MAITRI (1989) → Schirmacher Oasis, Antarctica; operational

MECHANISM
- BHARATI (2012) → Larsemann Hills, Antarctica; operational
- HIMADRI (2008) → Ny-Alesund, Svalbard; Arctic; HIMANSH → Spiti, HP

CONSEQUENCE / LIMIT
- CLOSE DISTINCTION: Arctic is mainly an ocean surrounded by continents while Antarctica is a high continent surrounded by ocean
- sea ice differs from land ice, tundra from ice cap, and polar desert refers to low precipitation rather than absence of frozen water.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Arctic is mainly an ocean surrounded by continents while Antarctica is a high continent surrounded by ocean; sea ice differs from land ice, tundra from ice cap, and polar desert refers to low precipitation rather than absence of frozen water.

10

STAGE 10 INDIA POLAR LEGAL FRAMEWORK

INDIA POLAR LEGAL FRAMEWORK · ANTARCTIC TREATY 1959 · INDIA ACCEDED 1983; CONSULTATIVE PARTY · INDIAN ANTARCTIC ACT 2022 · DOMESTIC LEGISLATION FOR POLAR ACTIVITIES · ARCTIC POLICY 2022 · SIX PILLARS; OBSERVER IN ARCTIC FORUM · RESEARCH FOCUS

ANTARCTIC TREATY 1959 → India acceded 1983; consultative party → INDIAN ANTARCTIC ACT 2022 → domestic legislation for polar activities → ARCTIC POLICY 2022 → six pillars; observer in Arctic forum → RESEARCH FOCUS → teleconnections: Arctic ice loss to Indian monsoon disruption → EVIDENCE LIMIT: Amplification, sea-ice loss, ice-sheet mass balance and governance claims require hemisphere, variable, baseline

CORE CLAIM
- ANTARCTIC TREATY 1959 → India acceded 1983; consultative party
- INDIAN ANTARCTIC ACT 2022 → domestic legislation for polar activities

MECHANISM
- ARCTIC POLICY 2022 → six pillars; observer in Arctic forum
- RESEARCH FOCUS → teleconnections: Arctic ice loss to Indian monsoon disruption

CONSEQUENCE / LIMIT
- EVIDENCE LIMIT: Amplification, sea-ice loss, ice-sheet mass balance and governance claims require hemisphere, variable, baseline and date
- Ladakh/cold deserts are useful process comparisons but not polar climates.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Amplification, sea-ice loss, ice-sheet mass balance and governance claims require hemisphere, variable, baseline and date; Ladakh/cold deserts are useful process comparisons but not polar climates.

11

STAGE 11 POLAR CLIMATE ANSWER SPINE

POLAR CLIMATE ANSWER SPINE · COLD DESERT NORTH OF ARCTIC CIRCLE; ICE-CAP VS TUNDRA · ARCTIC OCEAN VS ANTARCTIC CONTINENT; FLOATING VS GROUNDED ICE · ICE-ALBEDO FEEDBACK, PERMAFROST CARBON, FOUR TRANSMISSION CHANNELS · INDIA'S LADAKH ANALOGUE, POLAR RESEARCH, 2021 PYQ FOUR-CHANNEL SPINE

DEFINE → cold desert north of Arctic Circle; ice-cap vs tundra subdivisions → DISTINGUISH → Arctic ocean vs Antarctic continent; floating vs grounded ice → EXPLAIN → ice-albedo feedback, permafrost carbon, four transmission channels → QUALIFY → India's Ladakh analogue, polar research, 2021 PYQ four-channel spine

CORE CLAIM
- DEFINE → cold desert north of Arctic Circle; ice-cap vs tundra subdivisions
- DISTINGUISH → Arctic ocean vs Antarctic continent; floating vs grounded ice

MECHANISM
- EXPLAIN → ice-albedo feedback, permafrost carbon, four transmission channels
- QUALIFY → India's Ladakh analogue, polar research, 2021 PYQ four-channel spine

CONSEQUENCE / LIMIT
- DEFINE → cold desert north of Arctic Circle; ice-cap vs tundra subdivisions
- DISTINGUISH → Arctic ocean vs Antarctic continent; floating vs grounded ice

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALIFY → India's Ladakh analogue, polar research, 2021 PYQ four-channel spine.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · NCPOR (GOA) RUNS INDIA'S POLAR SCIENCE; INDIA ACCEDED TO · DAKSHIN GANGOTRI (1ST, 1983–90) · MAITRI (1989, SCHIRMACHER OASIS) · RESEARCH THEME · TELECONNECTIONS — ARCTIC SEA-ICE LOSS IS LINKED TO DISRUPTION · INDIA'S PRINCIPAL COLD-DESERT REGION; DISTINGUISH TRANS-HIMALAYAN KARAKORAM/LADAKH RANGES · COLD-DESERT BIOME SHARED WITH SIERRA NEVADA (USA) AND THE

OPTIONAL SOURCE DEPTH
- CAUTION: NCPOR (Goa) runs India's polar science; India acceded to the Antarctic Treaty (1983) and has
- CAUTION: Stations: Dakshin Gangotri (1st, 1983–90) → Maitri (1989, Schirmacher Oasis) → Bharati
- CAUTION: Research theme: teleconnections — Arctic sea-ice loss is linked to disruption of the Indian
- Ladakh = India's principal cold-desert region; distinguish Trans-Himalayan Karakoram/Ladakh ranges

ADVANCED DEBATE
- Cold-desert biome shared with Sierra Nevada (USA) and the Andes (Argentina) — light winter
- CAUTION: India's polar/high-altitude stations: Maitri, Bharati (Antarctic), Himadri (Arctic),
- CAUTION: Legal/policy: Antarctic Treaty 1959 (India 1983), Indian Antarctic Act 2022 , Arctic Policy
- NOT: Ladakh is arid because it is hot → arid due to rain-shadow + high altitude (a cold desert).

LEGACY / NUANCE
- NOT: Bharati is India's Arctic station → Bharati & Maitri are ANTARCTIC ; Himadri is the Arctic
- NOT: Dakshin Gangotri is operational → it was the first station (1983), later decommissioned/buried
- NOT: Ladakh gets its rain from the SW monsoon → mainly light winter snow from western disturbances .
- "The Ladakh cold desert is a fragile high-altitude ecosystem under climate and tourism stress." Discuss

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels